

[illegible]

No.	Task /Activity	Hazard and Consequence		Initial Risk			Mitigating Controls	Residual Risk		
	Sequential Steps	Hazard	Consequence	Probability	Consequence Severity	Risk Rating	Measures to Reduce Risk to ALARP	Probability	Consequence Severity	Risk Rating
1	Review TRA and TBT	Miscommunication	Personnel Injury / Environmental Pollution	1	2	2	Situational awareness and proper use of PPE. Hard hats, impact gloves, cotton gloves, latex gloves, steel toed boots, safety glasses, side shields and face shield	1	1	1
2	Prior to any task being undertaken ensure that all personnel are aware of the hazards involved.	Electrical shock, possible slips trips and falls. Pinch points poor body position. Hydraulic discharge and spills.	Personnel Injury / Environmental Pollution	2	3	6	All personnel must be aware of their surrounds and weather conditions. Use proper Manual Handling techniques, i.e., Always use proper body positioning, lift with legs, and obtain second personnel when loads are over 40lbs. Read all SDS on chemicals used, use PPE that is specified in SDS and be aware of where all spill kits are located.	2	2	4
Superintendents/Supervisors or trained personnel are to fill out form DO-HSE-FRM-062										
TMS										
3	De energizing TMS. Ensure that TMS grounding wheel is in the grounded position and a Lock is placed on grounding wheel.	Electrical shock, damage to equipment	Personnel Injury / Equipment Damage	3	4	12	Personnel are to make sure that red mark is in the ground location. Red light will indicate that TMS is grounded, megacon indicator will show grounded and an audible tone will sound. If any of these are not seen or heard check system. All personnel working on the task are to have their own lock on the system to avoid un-intentional energizing of the system	2	2	4
4	Ensure that TMS Power is in the off position and that a lock is in place.	Electrical shock	Personnel Injury / Equipment Damage	2	2	4	All personnel working on the task are to have their own lock on the system to avoid un-intentional energizing of the system. Test dead before any work is to be done on system.	1	1	2

ROV										
5	De energizing ROV power. Ensure that ROV-IPDS and HPU grounding wheel is in the grounded position and a Lock is placed on grounding wheel.	Electrical shock, damage to equipment	Personnel Injury / Equipment Damage	3	4	12	Personnel are to make sure that red mark is in the ground location. Red light will indicate that ROV IPDS and HPU is grounded, megacon indicator will show grounded and an audible tone will sound. If any of these are not seen or heard check system. All personnel working on the task are to have their own lock on the system to avoid un-intentional energizing of the system	2	2	4
6	Ensure that ROV IPDS and ROV HPU is in the off position and that a lock is in place on each one.	Electrical shock	Personnel Injury / Equipment Damage	2	2	4	All personnel working on the task are to have their own lock on the system to avoid un-intentional energizing of the system. Test dead before any work is to be done on system.	1	1	2
MAIN POWER										
7	De energizing Main Power to ROV/TMS. Ensure that steps 3 thru 6 have been performed if main power is being de energized. A lock is to be put on Main Power, and the main breaker should be turned off in the Power distribution panel.	Electrical shock	Personnel Injury / Equipment Damage	3	4	12	All personnel working on the task are to have their own lock on the system to avoid un-intentional energizing of the system. Test dead before any work is to be done on system.	2	2	4
MAIN VESSEL POWER										
8	Turning off main power to entire system. TS power to HPU	Electrical Shock	Personnel Injury / Equipment Damage	3	4	12	Contact bridge and engine room for vessel to turn off main power to system. Supv to LOTO in ER with ownership of LOTO key. Test dead before any work is to be done on system. Obtain Permit to work if required by vessel.	2	2	4

9	Reenergizing system	Electrical shock, damage to system	Personnel Injury / Equipment Damage	3	4	12	Before power is reenergized all equipment is to be checked to make sure that all boxes and/or junctions are closed, and nothing left exposed. Remove Lock-out locks from grounding wheels and main breaker. Personnel will be in constant communications with pilot as system is being tested. If main vessel power was de energized, then bridge and engine room must be contacted to reenergize system.	2	2	4
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TRA Approval	Name	Position	Signature	Date
By signing, the Approver confirms that the TRA has been reviewed and considers the task acceptable to proceed with residual risk meeting the acceptance criteria of DO Risk Matrix.				

*Reference DO-HSE-FRM-140 for DeepOcean Risk Assessment Matrix (RAM) and Guidewords